

Mathematics 1983

1. If M represents the median and D the mode of the measurements 5, 9, 3, 5, 8 then (M,D) is
 A. (6,5) B. (5,8) C. (5,7)
 D. (5,5) E. (7,5)
10. If $x + 2$ and $x - 1$ are factors of the expressions $lx + 2kx^2 + 24$, find the values of l and k
 A. $l = -6, k = -9$ B. $l = -2, k = 1$ C. $l = -2, k = -1$
 D. $l = 0, k = 1$ E. $l = 6, k = 0$

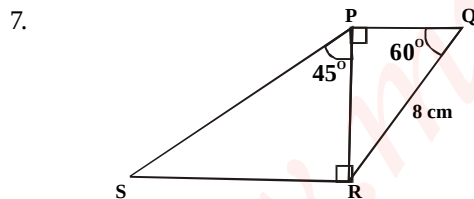
2. A construction company is owned by two partners X and Y and it is agreed that their profit will be divided in the ratio 4:5. at the end of the year. Y received #5,000 more than x. what is the total profit of the company for the year?
 A. #20,000.00 B. P'0#25,000.00 C. #30,000.00
 D. #15,000.003 E. #45,000.00

3. Given a regular hexagon, calculate each interior angle of the hexagon.
 A. 60° B. 30° C. 120°
 D. 45° E. 135°

4. Solve the following equations
 $4x - 3 = 3x + y = 2y + 5x - 12$
 A. $4x = 5, y = 2$ B. $x = 2, y = 5$ C. $x = -2, y = -5$
 D. $x = 5, y = -2$ E. $x = -5, y = -2$

5. If $x = 1$ is root of the equation $x^3 - 2x^2 - 5x + 6$, find the other roots
 A. -3 and 2 B. -2 and 2 C. 3 and -2
 D. 1 and 3 E. -3 and 1

6. If x is jointly proportional to the cube of y and the fourth power of z . In what ratio is x increased or decreased when y is halved and z is doubled?
 A. 4:1 increase B. 2:1 increase C. 1:4 decrease
 D. 1:1 no change E. 3:4 decrease



In the above figure $PQR = 60^\circ$, $QPR = 90^\circ$, $PRS = 90^\circ$, $RPS = 45^\circ$, $QR = 8\text{cm}$. Determine PS

- A. $2\sqrt{3}\text{cm}$ B. $4\sqrt{6}\text{cm}$ C. $2\sqrt{6}\text{cm}$
 D. $8\sqrt{6}\text{cm}$ E. 8cm

8. Given that $\cos z = L$, where z is an acute angle find an expression for $\frac{\cos z + \sec z}{\sec z + \tan z}$

- A. $\frac{1-L}{1+L}$ B. $\frac{L^2 - \sqrt{1-L^2}}{L^2 + L - 1}$ C. $\frac{-L - \sqrt{1-L}}{(L+1) + \sqrt{1-L^2}}$

- D. $\frac{\sqrt{L-1}}{(L+1) + \sqrt{1-L^2}}$ E. $\frac{L - (L^2 - 1)}{1 + \sqrt{1-L^2} + \sqrt{1-L^2}}$

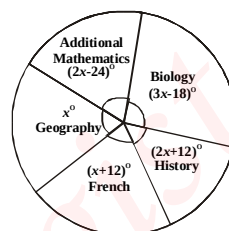
9. If $0.0000152 \times 0.00042 = A \times 10^8$, where $1 \leq A < 10$, find A and B.
 A. $A = 9, B = 6.38$ B. $A = 6.38, B = -9$ C. $A = 6.38, B = 9$
 D. $A = 6.38, B = -1$ E. $A = 6.38, B = 1$

11. Make T the subject of the equation

$$\frac{av}{1-v} = \sqrt[3]{\frac{2V+T}{a \cdot 2T}}$$

- A. $3av/(1-v)$ B. $2v(1-v)^2 - a^2v^2/2a^2v^2 - (1-v)^2$
 C. $2v(1-v)^2 + a^3v^2/2a^2v^2 + (1-v)^2$
 D. $2v(1-v)^2 - a^4v^3/2a^3v^3 - (1-v)^3$
 E. $2v(1-v)^3 - a^4v^3/2a^3v^3 + (1-v)^3$

- 12.



In a class of 60 pupils, the statistical distribution of the number of pupils offering Biology, History, French, Geography and Additional Mathematics is as shown in the pie chart above. How many pupils offer Additional Mathematics?

- A. 15 B. 10 C. 18
 D. 12 E. 28

13. The value of $(0.303)^3 - (0.02)^3$ is
 A. 0.019 B. 0.0019 C. 0.00019
 D. 0.000019 E. 0.000035

14. y varies partly as the square of x and y partly as the inverse of the square root of x . write down the expression for y if $y = 2$ when $x = 1$ and $y = 6$ when $x = 4$

- A. $y = \frac{10x^2}{31} + \frac{52}{31\sqrt{x}}$ B. $y = x^2 + \frac{1}{\sqrt{x}}$
 C. $y = x^2 + \frac{1}{x}$ D. $y = \frac{x^2 + 1}{31\sqrt{x}}$ E. $y = \frac{10(x^2 + 1)}{31(\sqrt{x})}$

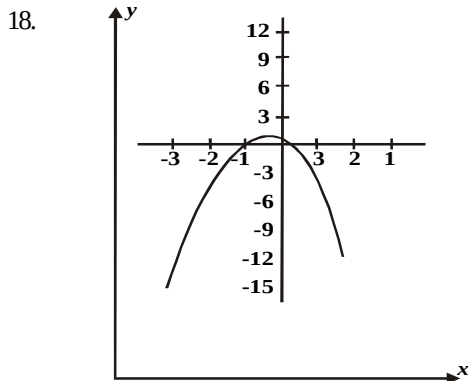
15. Simplify $(x-7)/(x^2-9) \cdot (x^2-3x)/(x^2-49)$
 A. $x/(x-3)(x+7)$ B. $(x+3)(x+7)/x$ C. $x/(x-3)(x-7)$
 D. $x/(x+3)(x+7)$ E. $x/(x+4)(x+7)$

16. The lengths of the sides of a right-angled triangle at $(3x+1)\text{cm}$, $(3x-1)\text{cm}$ and $x\text{cm}$.

- A. 2 B. 6 C. 18
 D. 12 E. 0

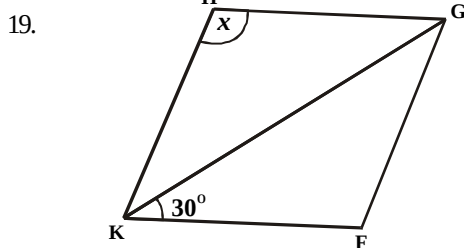
17. The scores of a set of a final year students in the first semester examination in a paper are 41,29,55,21,47,70,70,40,43,56,73,23,50,50. find the median of the scores.

- A. 47 B. $48\frac{1}{2}$ C. 50
 D. 48 E. 49



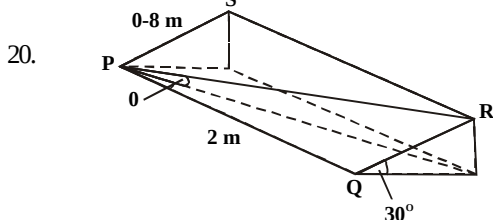
Which of the following equations represents the above graph?

- A. $y = 1 + 2x + 3x^2$ B. $y = 1 - 2x + 3x^2$ C. $y = 1 + 2x - 3x^2$
D. $y = 1 - 2x - 3x^2$ E. $y = 3x^2 + 2x - 1$



The above figure FGHK is a rhombus. What is the value of the angle x?

- A. 90° B. 30° C. 150°
D. 120° E. 60°



PQRS is a desk of dimensions 2m x 0.8m which is inclined at 30° to the horizontal. Find the inclination of the diagonal PR to the horizontal.

- A. $23^\circ 35'$ B. 30° C. $15^\circ 36'$
D. 10° E. $10^\circ 42'$

21. Find x if $(x_{\text{base } 4})^2 = 100 \text{ } 1000_{\text{base } 2}$
A. 6 B. 12 C. 100
D. 210 E. 110

22. Simplify $\log_{10} a^{1/2} + 1/4 \log_{10} a - 1/12 \log_{10} a^7$
A. 1 B. $7/6 \log_{10} a$ C. 0
D. 10 E. a

23. If w varies inversely as V and u varies directly as w^3 , find the relationship between u and V given that u = 1, when V = 2

- A. $u = 8V^3$ B. $u = 2\sqrt{V}$ C. $V = 8/u^2$
D. $V = 8u^2$ E. $U = 8\sqrt{v^3}$

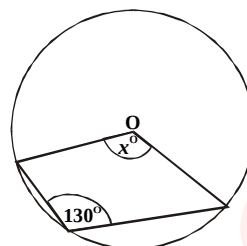
24. Solve the simultaneous equations for x
 $x^2 + y - 8 = 0$
 $y + 5x - 2 = 0$

- A. -28, 7 B. 6, -28 C. 6, -1
D. -1, 7 E. 3, 2

25. Find the missing value in the following table.

x	-2	-1	0	1	2	3
$y = x^3 - x + 3$		3	3	3	9	27

- A. -3 B. 3 C. -9
D. 13 E. 9



If O is the centre of the circle in the figure above. Find the value of x

- A. 50 B. 260 C. 100
D. 65 E. 130

27. Find the angle of the sectors representing each item in a pie chart of the following data. 6, 10, 14, 16, 26
A. $15^\circ, 25^\circ, 35^\circ, 40^\circ, 65^\circ$ B. $60^\circ, 100^\circ, 140^\circ, 160^\circ, 260^\circ$
C. $6^\circ, 10^\circ, 14^\circ, 16^\circ, 26^\circ$ D. $30^\circ, 50^\circ, 70^\circ, 80^\circ, 130^\circ$
E. None of the above

28. The scores of 16 students in a Mathematics test are 65, 65, 55, 60, 60, 65, 60, 70, 75, 70, 65, 70, 60, 65, 65, 70
What is the sum of the median and modal scores?
A. 125 B. 130 C. 140
D. 150 E. 137.5

29. The letters of the word MATRICULATION are cut and put into a box. One of the letter is drawn at random from the box. Find the probability of drawing a vowel.

- A. $2/13$ B. $5/13$ C. $6/13$
D. $8/13$ E. $4/13$

30. Correct each of the number 59.81789 and 0.0746829 to three significant figures and multiply them, giving your answer to three significant figures.
A. 4.46 B. 4.48 C. 4.47
D. 4.49 E. 4.50

31. If a rod of length 250cm is measured as 255cm longer in error, what is the percentage error in measurement?
A. 55 B. 10 C. 5
D. 4 E. 2

32. If $(2/3)m(3/4)n = 256/729$, find the values of m and n
A. $m = 4, n = 2$ B. $m = -4, n = -2$ C. $m = -4, n = 2$
D. $m = 4, n = -2$ E. $m = -2, n = 4$

33. Without using tables find the numerical value of $\log_7 49 + \log_7 (1/7)$
A. 1 B. 2 C. 3
D. 7 E. 0

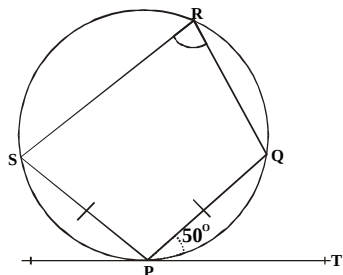
34. Factorize completely $81a^4 - 16b^4$

- A. $(3a + 2b)(2a - 3b)(9a^2 + 4b^2)$
 B. $(3a - 2b)(2a - 3b)(4a^2 - 9b^2)$
 C. $(3a - 2b)(3a - 2b)(9a^2 + 4b^2)$
 D. $(3a - 2b)(2a - 3b)(9a^2 + 4b^2)$
 E. $(3a - 2b)(2a - 3b)(9a^2 - 4b^2)$

35. One interior angle of a convex hexagon is 170° and each of the remaining interior angles is equal to x° . find x

- A. 120° B. 110° C. 105°
 D. 102° E. 100°

36. PQRS is a cyclic quadrilateral in which $PQ = PS$. PT is a tangent to the circle and PQ makes an angle 50° with the tangent as shown in the figure below. What is the size of QRS?



- A. 50° B. 40° C. 110°
 D. 80° E. 100°

37. A ship H leaves a port P and sails 30km due South. Then it sails 60km due west. What is the bearing of H from P?

- A. $26^\circ 34'$ B. $243^\circ 26'$ C. $116^\circ 34'$
 D. $63^\circ 26'$ E. 240°

38. In a sample survey of a university community the following table shows the percentage distribution of the number of members per household.

No of members per household	1	2	3	4	5	6	7	8	Total
Number of households	3	12	15	28	21	10	7	4	100

- A. 4 B. 3 C. 5
 D. 4.5 E. None

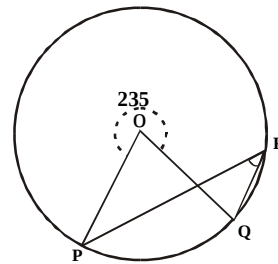
39. On a square paper of length 2.524375cm is inscribed a square diagram of length 0.524375. find the area of the paper not covered by the diagram correct to 3 significant figures.

- A. 6.00cm^2 B. 6.10cm^2 C. 6cm^2
 D. 6.09cm^2 E. 4.00cm^2

40. If $f(x) = \frac{1}{x-1} + \frac{x-1}{x^2-1}$ find $f(1-x)$

- A. $1/x + 1/(x+2)$ B. $x + 1/(2x-1)$
 C. $-1/x - 1/(x-2)$ D. $-1/x + 1/(x^2-1)$

41. In the figure below find PRQ

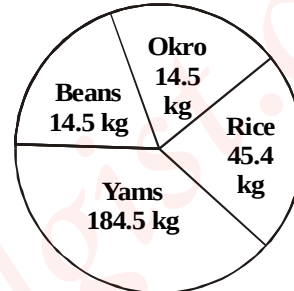


- A. $66\frac{1}{2}^\circ$ B. $62\frac{1}{2}^\circ$ C. 125°
 D. 105° E. 65°

42. Simplify $\sqrt{27a^9/8}$

- A. $9a^2/2$ B. $9a^3/2$ C. $2/3a^2$
 D. $2/3a^2$ E. $3a^3/2$

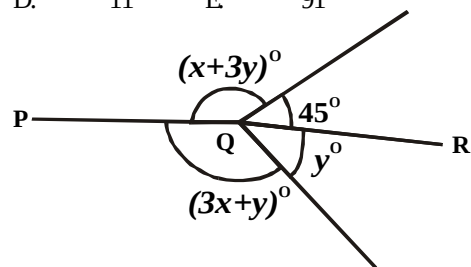
43.



The farm yields of four crops on a piece of land in Ondo are represented on the pie chart above. What is the angle of the sector occupied by Okro in the chart?

- A. $91\frac{1}{2}^\circ$ B. $19\frac{1}{3}^\circ$ C. $33\frac{1}{3}^\circ$
 D. 11° E. 91°

44.

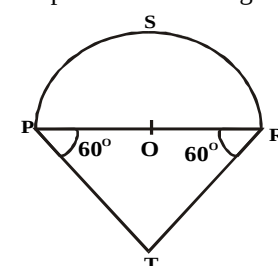


In the figure above, PQR is a straight line. Find the values of x and y

- A. $x = 22.5^\circ$ and $y = 33.75^\circ$
 B. $x = 15^\circ$ and $y = 52.5^\circ$
 C. $x = 22.5^\circ$ and $y = 45.0^\circ$
 D. $x = 56.25^\circ$ and $y = 11.5^\circ$
 E. $x = 18.^\circ$ and $y = 56.5^\circ$

45.

PQR is the diameter of a semicircle RSP with centre at Q and radius of length 3.5cm. if $\angle QPT = \angle QRT = 60^\circ$. Find the perimeter of the figure (PTRS $p = 22/7$)



- A. 25cm B. 18cm C. 36cm
 D. 20cm E. 25.5cm